

Teaching Students to Question the Machine: An AI Literacy Intervention Improves Students Regulation of LLM Use in a Science Task

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Abstract. The rapid adoption of generative artificial intelligence (GenAI) in schools raises concerns about students' uncritical reliance on its outputs. Effective use of large language models (LLMs) requires not only technical knowledge but also the ability to monitor, evaluate, and regulate one's interaction with the system, processes closely tied to metacognitive regulation. These skills are still developing in middle school, making students particularly vulnerable to over-trust and premature acceptance of AI outputs. Because classroom time and teacher training resources are constrained, there is a pressing need to develop and evaluate AI literacy interventions that can be implemented under realistic school conditions.

We report a controlled classroom study examining whether a two-hour AI literacy workshop improves students’ interaction strategies and quality of final answers (performance) in LLM-supported science problem solving. A total of 116 students (grades 8–9; ages 13–15) completed six science investigation tasks using a generative AI system. Two days prior, the intervention group attended the workshop, combining information about how LLMs work and fail with practical guidance on prompting and response evaluation; the control group received no training. Trained students showed less uncritical reliance on the system: they more often reformulated queries, asked follow-up questions, and more accurately judged response correctness, leading to better performance. In contrast, GenAI and metacognitive self-report scores did not predict performance, suggesting that effective use of generative AI depends less on self-reported measures and more on explicit training in interaction regulation.

Overall, the results show that brief, scalable AI literacy instruction can meaningfully improve how middle-school students use generative AI in school-like learning activities.

Keywords: Artificial intelligence in education · AI literacy · Generative AI · Self-regulation · Intervention

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** Equal supervision.

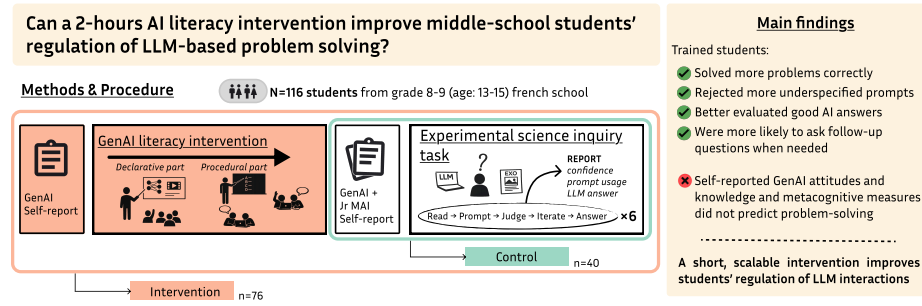


Fig. 1. Graphical abstract Study design and main findings. Grade 8–9 students (N=116) completed six science inquiry tasks with a GenAI system; the intervention group (n=76) received a two-hour AI literacy workshop (conceptual + procedural) two days before the task, whereas the control group did not (n=40). Trained students achieved higher final-answer scores and showed stronger interaction regulation (more rejection of underspecified prompts, more follow-up questions when needed, and more accurate correctness judgments), while neither GenAI nor metacognitive self-report scores predicted performance.

1 Introduction

Generative artificial intelligence (GenAI) systems are rapidly entering students' everyday school and learning practices. Recent surveys show rapid uptake of GenAI tools across age groups, from undergraduates to teenagers, including high rates in secondary science, technology, engineering, and mathematics (STEM) contexts [10,29,33]. GenAI chatbots (such as ChatGPT, Gemini, and Mistral-like assistants) are increasingly used for activities ranging from generating explanations to solving subject-specific problems [42]. This shift creates educational opportunities, as these systems can support personalization and adaptive learning [20,26,19], but it also raises well-documented risks.

A central challenge is that the outputs of large language models (LLMs) are often fluent and persuasive even when incomplete or incorrect. They can hallucinate facts, reproduce biases, or comply with misleading instructions, which can encourage over-trust and premature acceptance [12,39,32,15]. Another concern with potentially stronger long-term consequences is that students may delegate key reasoning steps to the system, reducing effortful processing and potentially weakening learning and critical thinking [11,30,41], an effect known as cognitive offloading [27]. In this context, the educational value of GenAI depends not only on what models generate, but on how students use them: how they formulate queries, inspect answers, and decide whether to accept, revise, or reject model outputs [35,17,2]. These concerns motivate growing interest in AI literacy as a target for educational interventions. Recent frameworks typically distinguish (i) foundational understanding of GenAI systems, (ii) practical communication and evaluation skills, and (iii) broader psychological and social/ethical dimensions [7,18,38]. Across frameworks, a recurring idea is that effective use re-

quires both conceptual knowledge (e.g., what LLMs are optimized for and why they fail) and procedural skills to apply that knowledge during real interactions. From this perspective, effective LLM-supported problem solving requires learners to plan queries based on informational needs, monitor response adequacy and plausibility, and select appropriate control actions (e.g., reformulate queries, ask follow-up questions, reject outputs). This regulation process has been linked to metacognitive monitoring and control in technology-mediated learning [31,37].

In practice, many learners do not reliably apply effective interaction strategies without explicit support. For example, university students often struggle to adequately assess LLM-generated responses and can sometimes perform worse when revising AI outputs than when working independently [14]. This is particularly concerning for adolescents, whose critical evaluation and self-regulatory skills are still developing and may be further challenged by persuasive LLM outputs [30,41,4]. Consistent with this, Abdelghani et al. [1] found that middle-school students struggled to select effective prompts or challenge unsatisfactory answers; notably, performance correlated positively with metacognitive regulation but negatively with self-reported AI expertise, suggesting that students significantly overestimate their own AI literacy.

These findings motivate AI literacy interventions that support interaction regulation by providing concrete heuristics for when and how to intervene during an exchange. Despite these concerns and the potential for AI literacy interventions, controlled classroom evidence for AI literacy interventions, particularly at the middle school level, remains limited. Most empirical work focuses on higher education, with comparatively fewer studies in K–12 settings [13], and existing school-based interventions are often constrained by small samples or missing control groups [5,25]. Moreover, school constraints make it critical to evaluate brief, instructionally lightweight interventions that can realistically be implemented in classrooms, as highly intensive AI literacy programs are unlikely to scale.

To address this gap, we conducted a controlled classroom study in a French middle school ($N = 116$, grades 8–9, ages 13–15) to evaluate whether a two-hour AI literacy workshop improves students’ regulation and performance during LLM-supported science problem solving. Two days after the workshop, students completed six investigation-style science tasks with access to a GenAI system; a control group completed the same session without prior training. While other interaction modes exist (e.g., collaborative writing, extended projects), this investigation-based problem-solving format allows systematic observation of key regulation behaviors (prompt formulation, output evaluation, iterative refinement) while maintaining experimental control and reflecting typical classroom and homework activities. The workshop combined conceptual content on how LLMs work and why they can fail with procedural guidance for interacting with and evaluating GenAI outputs. Importantly, rather than teaching a specific prompt-engineering technique or performance with a particular model, the intervention targeted general-purpose interaction principles intended to transfer across GenAI systems. We assessed both interaction regulation and final performance (students’ exercise scores) to determine whether changes in interac-

tion strategies improve outcomes. Regulation was measured behaviorally during each task, including whether students accept or replace a suggested prompt, whether they ask follow-up questions after an initial response, and how accurately they judge the correctness of the model’s answer. Figure 1 provides a graphical overview of the study design and main findings.

Contributions. This paper makes three main contributions:

- We provide controlled classroom evidence that a brief AI literacy workshop improves middle-school students’ performance in LLM-supported science problem solving.
- We show behavioral changes in students’ regulation of GenAI interactions, including prompt replacement, follow-up querying, and more accurate correctness judgments.
- We demonstrate measurable benefits under realistic school constraints using transferable interaction principles, rather than platform-specific prompt templates.

2 Method

2.1 Participants

The study was conducted in one French middle school during regular class hours. A total of 162 students (grades 8 and 9; typical age range 13–15 years) initially participated. Due to end-of-year constraints, 27 students who attended the workshop were absent during the test phase and were excluded. An additional 19 students did not complete all exercises and were also excluded. The final sample therefore comprised 116 students, divided into a control group ($n = 40$; 24 grade 8, 16 grade 9; 60% vs. 40%) and an intervention group ($n = 76$; 50 grade 8, 26 grade 9; 66% vs. 34%). We did not collect gender or race/ethnicity due to data-anonymization and feasibility constraints in a school context; consequently, we cannot test differential effects across demographic groups and generalizability beyond this context remains to be established. Both groups were drawn from the same school and tested during the same period.

Within the intervention group, four students provided inconsistent identifiers between the pre- and post-intervention GenAI questionnaires. These students were retained for all main behavioral analyses, but excluded from questionnaire pre-post comparisons. Accordingly, questionnaire analyses were conducted on $n = 40$ control students and $n = 72$ intervention students.

2.2 Procedure

Control students completed a 90-min exercise session, individually, on science problems using a LLM as a support tool. Intervention students completed the same session, preceded two days earlier by a 2-hour workshop (details below). A timeline and classroom procedure are provided in the Supplementary Material (SM Figs. S1–S2).

Self-reported measures. All students completed two questionnaires prior to the exercise phase. The first was a GenAI questionnaire adapted from Bernabei et al. [3] following Abdelghani et al. [1], covering six dimensions (Attitude, Trust, Social influence, Fairness & ethics, Usefulness, Effort & ease of use; 24 items; 4-point Likert; maximum score = 96). Because this adapted version has not been formally validated for this age group, GenAI self-reports were treated as secondary and interpreted cautiously. The second was the Jr. MAI [16], a measure developed for middle-school populations assessing metacognitive knowledge and regulation of cognition (18 items; 5-point Likert; maximum score = 90). We refer to these measures as GenAI self-reports and metacognitive self-reports throughout the paper; full item wording and scoring details are provided in SM S2. Questionnaires served two purposes. First, GenAI and metacognitive self-report scores were used as individual-difference predictors to test whether students’ self-perceptions were associated with task performance. Second, in the intervention group only, GenAI self-reports were collected both before and after the workshop as a secondary outcome to assess whether the workshop changed these self-reported dimensions.

Pedagogical intervention. The intervention was a two-hour, classroom-feasible AI literacy workshop designed to strengthen conceptual understanding of how LLM-based GenAI systems work and why they can fail, and procedural skills for interacting with and evaluating GenAI outputs during learning tasks [7,18,38]. The workshop combined short instructor-led segments with guided practice and peer discussion, following an inquiry-based instructional cycle [24]. The conceptual component covered core mechanisms (e.g., text generation and alignment) and limitations relevant to school use (e.g., hallucinations, bias, and sycophancy effects). The procedural component introduced transferable interaction heuristics (e.g., specifying context/constraints, requesting step-by-step explanations, asking targeted follow-up questions, and checking answers for plausibility and consistency). Workshop materials and activity descriptions are provided in the Supplementary Material (SM S3).

Task description. The exercise session was based on the same science problem-solving tasks described in Abdelghani et al. (2025). The experimental task consisted of six science exercises to be completed within 90 minutes (Fig. 2). Each student received a randomly ordered set of six exercises, selected from a pool of twelve science tasks, to minimize collaborative solving between students. Each exercise included a single suggested prompt, which could be either efficient (i.e., containing the key contextual elements needed for the exercise) or inefficient (i.e., intentionally underspecified and likely to elicit generic or incomplete responses). Prompt quality was pseudorandomized such that each student encountered three well-specified and three underspecified prompts. Students were not informed about this manipulation, and exercises were provided on paper to prevent direct copy-paste of the problem statement into the chatbot interface. Students worked individually at a computer with two browser tabs open:

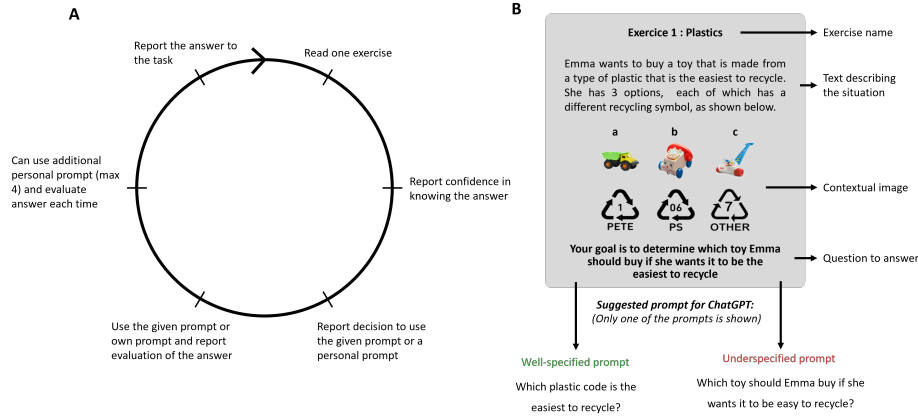


Fig. 2. Student-LLM workflow and exercise structure. (A) For each exercise, students rated confidence (3-point), decided whether to use the suggested prompt, iteratively queried and evaluated the LLM (3-point; optional follow-up), and wrote a final answer; repeated across six exercises. (B) Example sheet and prompt manipulation. The suggested prompt was either well-specified or underspecified (one shown per exercise).

an LLM-based chatbot and a web questionnaire guiding the procedure. They primarily used GPT-4o mini *via* DuckDuckAI; when this interface failed, some used other public chatbot interfaces running GPT-4o, GPT-4.1-nano, or GPT-3.5-turbo (robustness checks: SM S4.2–S4.4). For each exercise, students could interact iteratively with the LLM and evaluate its responses before producing a final written answer in their own words.

LLM-as-judge for prompt validation and student-answer scoring. We used GPT-4o as an LLM-as-judge in two parts of the study: (i) to validate the prompt manipulation by rating the quality of model answers elicited by well-specified versus underspecified prompts, and (ii) to score the quality of students’ final written answers on a 0–2 rubric (0 = incorrect, 1 = partially correct, 2 = correct). In both cases, we followed the same validation procedure: we first evaluated the annotation setup (model and prompt) on a random $\approx 10\%$ subset independently rated by human annotators, and applied GPT-4o to the full dataset only after meeting predefined reliability criteria (Krippendorff’s α and the advantage-probability test; see SM S4). Per-student performance was computed by summing the six exercise scores and converting the total to a 0–20 scale.

3 Results

3.1 The intervention improves task performance

Score distributions differed between the intervention and control groups, with higher performance (final answer validity; 0–20 scale) observed in the intervention group ($M = 11.38$, $SD = 3.87$, $n = 76$ vs. $M = 10.29$, $SD = 3.04$, $n = 40$; Mann–Whitney $U = 1871$, $p = .040$, $r = .19$; Fig. 3A). Notably, absolute performance remained modest in both groups, despite access to a GenAI system; however, maximum scores were high (up to 18.3/20), indicating that the tasks were feasible under the current scoring and time constraints (see distributions in SM S5.1). To test whether this advantage could be explained by students’ selectivity in accepting well-specified versus underspecified suggested prompts, we computed signal detection indices. Sensitivity (d') was slightly higher in the intervention group than in the control group, although the difference did not reach significance (see distribution in SM S5.2), and response bias (β) did not differ between groups. Across participants, both indices significantly predicted exercise scores: higher sensitivity ($\beta = 1.76$, $p < .001$, 95% CI [1.06, 2.46]) and a more conservative response bias ($\beta = 1.65$, $p = .021$, 95% CI [0.25, 3.05]) were associated with better performance (Fig. 3B). However, the group $\times d'$ interaction did not reach statistical significance ($\beta = 1.36$, $p = .072$, see distribution in SM S5.2). Exploratory within-group regression analyses nonetheless revealed that both d' and β significantly predicted performance in the intervention group, whereas no reliable associations were observed in the control group. Once d' and β were included in the regression model, the direct group effect was no longer significant, consistent with the interpretation that the intervention enhanced the strategic use of prompt discrimination (teaching them what to do when they identified an underspecified prompt, e.g., reformulating or asking follow-up questions) rather than improving their discrimination ability itself.

3.2 The intervention improves regulation of interaction

A generalized linear model (GLM) predicting prompt acceptance (Fig. 3C) showed that trained students accepted fewer suggested prompts overall ($p = .018$). Students who were more confident in knowing the answer (before prompting the GenAI chatbot) were less likely to accept the prompt ($p = .002$), and prompt validity increased acceptance ($p = .040$), whereas the condition $\times$ validity interaction was not significant ($p = .224$). However, follow-up contrasts indicated a group difference for underspecified prompts: trained students were less likely than controls to accept underspecified prompts (51.5% vs. 66.7%, $p = .044$), with no group difference for well-specified prompts.

Rejecting underspecified prompts improved performance in both groups, whereas rejecting well-specified prompts reduced performance in the control group but not in the intervention group, yielding a significant group difference in the reject–accept contrast ($p = .009$; Fig. 4A). This suggests that the intervention helped students replace even well-specified prompts without a reliable performance penalty.

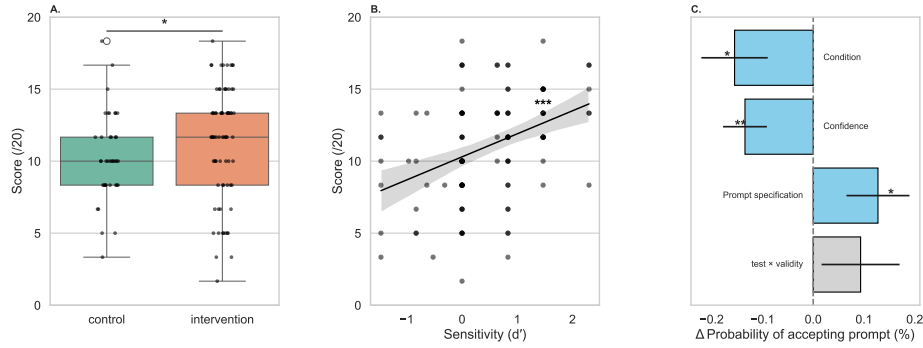


Fig. 3. Workshop effects on performance and prompt acceptance. (A) Final-answer performance (0–20) by group. Intervention students achieved higher final-answer scores. **(B)** Relationship between performance and prompt discrimination sensitivity (d'). Performance increased with prompt discrimination. **(C)** Predictors of accepting the suggested prompt (GLM).

Students in both groups were more accurate at judging the correctness of AI responses when prompts were well-specified rather than underspecified; however, this benefit was significantly larger in the intervention group than in the control group ($\times$ condition interaction, $p = .012$; Fig. 4B). They were also much more likely to ask follow-up questions when accepted initial prompts were underspecified (59.2% vs. 27.9%, $p < 0.001$; Fig. 4C), with no group difference for well-specified prompts. Crucially, in the intervention group this behavior translated into higher-quality answers (approximately +5 points out of 20), whereas no such benefit was observed in the control group (interaction $p = .038$).

Together, these results show that the intervention improved students' ability to evaluate prompt adequacy, assess output correctness, and adapt their interaction when initial responses were insufficient. Complete analyses are provided in the SM S5.3–S5.5.

3.3 No effect of the workshop on GenAI self-report scores

In this section, we tested whether the workshop changed students' GenAI self-report scores.

A one-way MANOVA on pre-test GenAI self-report dimensions revealed no overall difference between the intervention and control groups (*Pillai's trace* = 0.083, $F(6, 107) = 1.61$, $p = .151$). Follow-up univariate comparisons for each dimension, with FDR correction, showed no significant group differences (all $p_{\text{FDR}} \geq .189$), indicating that the groups were comparable at baseline.

A repeated-measures MANOVA revealed no overall effect of the intervention on the GenAI self-report dimensions in the intervention group (*Pillai's trace* = 0.116, $F(6, 62) = 1.36$, $p = .247$). Follow-up univariate paired comparisons for each dimension, with FDR correction, revealed no significant pre–post differences

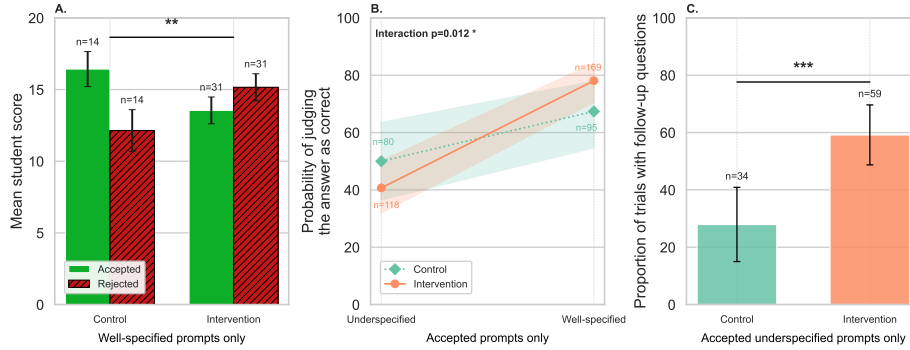


Fig. 4. Effects of the intervention on prompt replacement, follow-up behavior, and correctness judgments. (A) Within-student performance by accept vs. reject on well-specified prompts. Rejecting well-specified prompts was more costly in the control group. (B) Probability of judging an AI answer as correct (GLM; 95% CI) by prompt specificity. Trained students made more accurate correctness judgments. (C) Follow-up questioning after accepted underspecified prompts by group. Trained students asked more follow-up questions after underspecified prompts.

(all $p_{FDR} \geq .053$), although Attitude and Fairness & ethics showed small, non-robust increases from pre to post (Attitude: $p = .018$, $p_{FDR} = .054$; Fairness & ethics: $p = .011$, $p_{FDR} = .054$).

3.4 Self-report measures (GenAI and metacognition) do not predict performance

In this section, we treated GenAI and metacognitive self-report scores as individual-difference predictors to test whether they explained variability in task performance. Across all students, GenAI self-report scores were not significantly correlated with performance ($r = 0.01$, $p = .88$, $R^2 < .001$, $n = 116$). Within the intervention group, pre-post changes in GenAI self-report scores did not predict students' sensitivity (d') to prompt specificity across subscales ($R^2 = .12$, all $p > .05$). Metacognitive self-report scores likewise showed no significant association with exercise scores ($r = 0.04$, $p = .65$, $R^2 = .002$, $n = 116$). Additional analyses of self-report measures are provided in SM S5.6.

4 Discussion

This study provides controlled classroom evidence that a brief, classroom-feasible AI literacy workshop can improve middle-school students' GenAI-supported science problem solving under realistic school constraints. We observed (i) a modest but reliable improvement in performance (higher-quality final written answers) and (ii) converging process-level changes consistent with stronger interaction regulation. Trained students rejected underspecified prompts more often and

could replace suggested prompts without a performance penalty; they also evaluated outputs more accurately and asked follow-up questions more often when initial responses were insufficient. Although these behaviors relate conceptually to metacognitive monitoring and control, neither GenAI nor metacognitive self-report scores predicted performance, suggesting a gap between perceived competence and effective regulation. Absolute performance nonetheless remained modest in both groups (means around 11–12/20), suggesting that LLM access alone is insufficient for producing high-quality final answers. One likely contributor is that students had to extract and synthesize relevant information into a brief response in their own words, which is challenging at this age even when the source contains relevant content [36]. Because the activity was low-stakes (not graded), reduced motivation may also have contributed [8]; importantly, these constraints applied to both groups and therefore do not undermine the between-group effect.

4.1 The intervention supported regulation at two levels

The workshop provided brief conceptual input and procedural guidance for interacting with GenAI and evaluating outputs. This design targets an interaction regulation loop where learners plan queries, monitor response adequacy, and select control actions. We distinguished behaviors at the prompt and response stages. At the prompt level, trained students became more selective, rejecting underspecified prompts more often and replacing well-specified prompts without performance cost, suggesting greater agency in treating suggested prompts as starting points rather than ready-made solutions. At the response level, trained students monitored outputs more effectively, judging correctness more accurately after well-specified prompts and asking follow-up questions when initial answers were inadequate, indicating that they treated first responses as provisional rather than final. Overall, the workshop strengthened regulation of GenAI-supported problem solving at both stages.

4.2 From discrimination to strategy

The workshop’s benefit appears to reflect improved responses to detected problems more than improved detection itself. Although trained students rejected underspecified prompts more often, discrimination sensitivity increased only slightly and did not differ reliably between groups. Yet both sensitivity and response bias predicted performance, with the discrimination–performance relationship tending to be stronger in the intervention group. This pattern supports a monitoring-to-control account: identifying an underspecified prompt or incomplete answer improves outcomes only when it triggers effective control actions. The workshop seems to have strengthened this coupling between monitoring signals and appropriate action, suggesting that what mattered was less knowing how to prompt and more knowing how to regulate an inquiry.

4.3 Self-reported competence does not match effective use

Students’ GenAI self-reports were unrelated to performance: neither overall questionnaire scores nor pre–post changes predicted task success. Self-reported metacognitive awareness also did not predict performance. These null associations should be interpreted cautiously, particularly for GenAI self-reports, which have not been formally validated for this age group. While this may seem at odds with the metacognitive demands of LLM interaction [31], recent work suggests that offline self-report measures correlate weakly with online regulation during task performance, whereas behavioral indicators of monitoring and control better predict learning outcomes [28]. This aligns with evidence that self-reports can overestimate competence [9], with miscalibration often stronger in younger individuals for digital skills [23]. In this context, feeling confident with GenAI is not equivalent to using it effectively, underscoring the value of process- and performance-based measures when evaluating AI literacy interventions.

4.4 Implications for AI literacy interventions in middle school

Our results extend Abdelghani et al. [1] by showing that, without guidance, middle-school students struggle to use LLMs effectively. The control group replicated these difficulties in an independent cohort, consistent with evidence that non-experts often struggle to design robust prompts without structured support [40]. Although children can ask informative questions and evaluate explanations in non-GenAI contexts [21,22], these skills may be less reliably deployed in GenAI-mediated inquiry, potentially due to over-reliance on confident-looking outputs [41]. Encouragingly, even a brief intervention shifted interaction strategies and improved performance.

More broadly, meta-analytic work suggests that LLM use can improve performance, but controlled evidence in middle school remains scarce. Wang and Fan’s meta-analysis [34] found only one middle-school study with small or non-significant effects [6]. Many interventions rely on extended exposure, whereas our findings suggest that explicit instruction can accelerate acquisition of regulation strategies. Importantly, our intervention emphasized general principles about how LLMs work, why they fail, and how to regulate interactions iteratively—not platform-specific techniques. This focus may be particularly important for younger learners whose metacognitive regulation skills are developing and unlikely to emerge through unguided use. Because these regulations are not tied to a single platform, they may transfer to other information-seeking contexts (e.g., web search, textbooks, peer collaboration), though such transfer remains to be tested.

4.5 Limitations and directions for future research

Several limitations should be noted. First, the test occurred only two days after the workshop, so the durability of behavioral change is unknown; future studies

should assess longer-term outcomes and examine repeated instructional interventions rather than a one-time workshop. Second, we focused on middle-school science problem solving; generalization to other subjects and GenAI-supported learning contexts remains to be tested. Third, we did not analyze linguistic properties of prompts or outputs; fine-grained analyses could complement our behavioral measures. Fourth, the sample came from a single school and no demographic data were collected, limiting generalizability and preventing subgroup analyses; replication across diverse contexts is needed. Finally, using multiple LLM backends introduces potential variability in system behavior. However, this heterogeneity reflects ecological conditions, and our prompt-validation procedure confirmed that the experimental manipulation was robust across models.

Ethics and data protection. The study was approved by an institutional ethics committee (details withheld for double-blind review). In accordance with this approval, no personal data were collected. Participation required prior parent/guardian consent and student assent. No demographic information was collected, no data could be linked to individual identities, and no personal account or login was required to access the LLM interface; students were instructed not to enter any personal information.

Supplementary materials. SM containing additional analyses, questionnaires, and workshop resources is available at an anonymized OSF repository: https://osf.io/fyu7c/overview?view_only=cd809b8f5f6948c9a2536c824b46c56b.

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